

Tema 11

Preguntas Tipler

Campo Magnetico debido al movimiento de CP

Problema 13

$$\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{q \cdot |\vec{v} \times \hat{r}|}{r^2}$$

a) $\hat{r} = -\vec{j}$

$$B = \frac{\mu_0}{4\pi} \cdot \frac{12 \cdot 10^{-6}}{(2)^2} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 30 & 0 & 0 \\ 0 & -1 & 0 \end{vmatrix} \Rightarrow \vec{B} = \frac{4\pi \cdot 10^{-7}}{4\pi} \cdot \frac{12 \cdot 10^{-6}}{4} \cdot 30 \vec{k}$$

$$\vec{B} = -9 \cdot 10^{-12} \vec{k} \text{ T}$$

b) $\hat{r} = -\vec{j} \rightarrow B = \frac{\mu_0}{4\pi} \cdot \frac{12 \cdot 10^{-6}}{(2)^2} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 30 & 0 & 0 \\ 0 & -1 & 0 \end{vmatrix} = -3,6 \cdot 10^{-11} \vec{k} \text{ T}$

c) $\hat{r} = \vec{j} \rightarrow B = \frac{\mu_0}{4\pi} \cdot \frac{12 \cdot 10^{-6}}{(2)^2} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 30 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 3,6 \cdot 10^{-11} \vec{k} \text{ T}$

d) $\hat{r} = \vec{j} \rightarrow B = 9 \cdot 10^{-12} \vec{k} \text{ T}$

Problema 17

$$F_e = \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{b^2} \quad F_m = \frac{\mu_0}{4\pi} \cdot \frac{q^2 \cdot v^2}{b^2}$$

$$\frac{F_e}{F_m} = \frac{\frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{b^2}}{\frac{\mu_0}{4\pi} \cdot \frac{q^2 \cdot v^2}{b^2}} = \frac{q^2 \cdot 4\pi b^2}{\mu_0 q^2 \cdot v^2 \cdot 4\pi \epsilon_0} = \frac{1}{\mu_0 \epsilon_0 v^2} = \frac{c^2}{v^2}$$

Ley de Biot y Savart

Problema 19

$$I = 2A \quad l = 2 \cdot 10^{-3}m \quad r = \sqrt{3^2 + 4^2} = 5m \quad \hat{r} = \frac{3}{5}\vec{j} + \frac{4}{5}\vec{k}$$

$$B = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot dl \times \hat{r}}{r^2} = 2 \cdot 10^{-7} \cdot \frac{2}{25} \cdot \begin{vmatrix} i & j & k \\ 0 & 0 & 2 \cdot 10^{-3} \\ 0 & \frac{3}{5} & \frac{4}{5} \end{vmatrix} = -9,6 \cdot 10^{-12} T$$

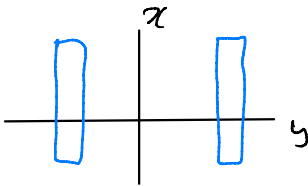
Problema 20

$$a) r = \sqrt{2^2 + 4^2} = \sqrt{20} = 2\sqrt{5} \quad \hat{r} = \frac{2}{2\sqrt{5}}\vec{i} + \frac{4}{2\sqrt{5}}\vec{j}$$

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2}{20} \cdot \begin{vmatrix} i & j & k \\ 0 & 0 & 2 \cdot 10^{-3} \\ \frac{2}{\sqrt{5}} & \frac{4}{\sqrt{5}} & 0 \end{vmatrix} = 8,94 \cdot 10^{-12} \vec{j} T$$

$$b) B = 8,94 \cdot 10^{-12} \vec{j} T$$

Campo Magnetico por lineas de corriente



Problema 25

$$B = \frac{\mu_0 I}{2\pi r} \quad I = 20$$

$$a) B_T = B_1 + B_2$$

$$B_T = \frac{\mu_0 I}{2\pi \cdot 3 \cdot 10^{-2}} - \frac{\mu_0 I}{2\pi \cdot 9 \cdot 10^{-2}} = \frac{3\mu_0 I}{2\pi \cdot 10^{-2}} - \frac{\mu_0 I}{2\pi \cdot 9 \cdot 10^{-2}} = \frac{2\mu_0 I}{2\pi \cdot 9 \cdot 10^{-2}}$$

$$B_T = 2 \cdot 10^{-7} \cdot \frac{2 \cdot 20}{9 \cdot 10^{-2}} = -8,9 \cdot 10^{-5} T$$

$$b) B_T = B_1 + B_2 = \frac{\mu_0 I}{2\pi \cdot 6} - \frac{\mu_0 I}{2\pi \cdot 6} = 0 T$$

$$c) B_T = \frac{\mu_0 I}{2\pi \cdot 9} - \frac{\mu_0 I}{2\pi \cdot 3} = 8,9 \cdot 10^{-5} T$$

$$d) B_T = \frac{\mu_0 I}{2\pi \cdot 15 \cdot 10^{-2}} + \frac{\mu_0 I}{2\pi \cdot 3} = \frac{6\mu_0 I}{2\pi \cdot 15 \cdot 10^{-2}} = 2 \cdot 10^{-7} \cdot \frac{6 \cdot 20}{15 \cdot 10^{-2}} = -160 \cdot 10^{-4} T$$

